

Sovereign credit spreads, banking fragility, and global factors[☆]Anusha Chari^a, Felipe Garcés^b, Juan Francisco Martínez^c, Patricio Valenzuela^{d,*}^a University of North Carolina at Chapel Hill, United States of America^b Central Bank of Chile, Chile^c Itaú Bank, Chile^d Universidad de los Andes, Chile

ARTICLE INFO

JEL classification:

E43

E44

F30

G12

G15

Keywords:

Banks

Credit ratings

Credit risk

Emerging economies

Global factors

ABSTRACT

This study explores the relationship between sovereign credit risk, banking fragility, and global financial factors in a large panel database of emerging market economies. To measure banking fragility, we construct a novel model-based semi-parametric metric (JLoss) that computes the expected joint loss of the banking sector in each country conditional on a country-level systemic event. Our metric of banking fragility is positively associated with sovereign credit spreads, after controlling for the standard determinants of sovereign credit risk, a comprehensive set of measures of systemic risk, and country and time fixed effects. The results additionally indicate that countries with more fragile banking sectors are more exposed to global (exogenous) financial factors than those with more resilient banking sectors. These findings underscore that regulators must ensure the stability of the banking sector to improve governments' borrowing costs in international debt markets.

1. Motivation

The global financial crisis of 2008–09 and the European debt crisis, which were characterized by large losses in the banking sector, affected international debt markets severely. They produced a significant deterioration of sovereign credit spreads with the greater expectation of public support for distressed banks (Mody and Sandri, 2012). Despite a rich body of research on the drivers of sovereign credit risk, a better understanding of the factors influencing sovereign risk and of how these factors can be properly measured in both advanced and emerging economies is of key importance for several reasons. Sovereign credit risk is not only a key determinant of governments' borrowing costs, but also remains a significant determinant of the cost of debt capital for the private sector (Cavallo and Valenzuela, 2010; Borensztein et al., 2013). Moreover, sovereign credit risk directly influences the ability of investors to diversify the risk of global debt portfolios and plays a crucial role in determining capital flows across countries (Longstaff et al., 2011).

The literature has recently emphasized that the primary factors that affect sovereign credit risk are macroeconomic fundamentals, global

factors, and banking fragility, which have generally been treated as independent determinants of sovereign credit risk. Although macroeconomic fundamentals have substantial explanatory power for sovereign credit spreads in emerging economies (Hilscher and Nosbusch, 2010), sovereign credit risk appears to be mainly driven by global financial factors (González-Rozada and Yeyati, 2008; Longstaff et al., 2011). Banking fragility also seems to influence governments' indebtedness and credit risk. Greater banking-sector fragility predicts larger bank bailouts, larger public debt, and higher sovereign credit risk (Acharya et al., 2014; Kallestrup et al., 2016; Farhi and Tirole, 2012). This relationship between bank risk and sovereign risk is particularly strong during periods of financial distress (Fratzscher and Rieth, 2019). Finally, recent empirical evidence also suggests systemic sovereign risk has its roots in financial markets rather than in macroeconomic fundamentals (Dieckmann and Plank, 2012; Ang and Longstaff, 2013). Specifically, Dieckmann and Plank (2012) show the state of the domestic financial market and the state of the global financial system have strong explanatory power for the evolution of sovereign spreads, and that the magnitude of the effect is shaped by the importance of the domestic financial system pre-crisis.

[☆] We wish to thank Charles Goodhart, Dimitrios Tsomocos, Jose Vicente Martinez, Oren Sussman, and participants at the First Conference on Financial Stability and Sustainability for insightful comments on an earlier version of this paper. Valenzuela acknowledges financial support from Fondecyt Project, Chile #1200070 and the Institute for Research in Market Imperfections and Public Policy, MIPP (ICS13002 ANID).

* Corresponding author.

E-mail address: pvalenzuela2@miuandes.cl (P. Valenzuela).

Using a novel model-based semi-parametric metric (JLoss) that computes the expected joint loss of the banking sector in the event of a large financial meltdown, in this study, we explore the relationship between sovereign credit spreads, banking fragility, and global financial factors. We study this relationship in a panel data set that covers 19 emerging market economies from 2000:Q1 to 2021:Q4. Consistent with the idea that our metric (JLoss) can be understood as the direct cost of bailing out the whole banking sector, and with recent evidence that shows sovereign spreads increased in the eurozone with the greater expectation of public support for distressed banks (Mody and Sandri, 2012), our results indicate our metric of banking fragility is positively associated with sovereign credit spreads. The results additionally indicate countries with more fragile banking sectors are more exposed to the influence of global (exogenous) financial factors related to market volatility, risk-free interest rates, risk premiums, and aggregate liquidity. Our results are statistically significant and economically meaningful, even after controlling for the standard determinants of sovereign credit risk, a comprehensive set of systemic risk measures, and country and time fixed effects. Our results are also robust during periods of financial distress and periods of financial stability as well as to controlling for additional heterogeneous effects of global financial factors on sovereign spreads. These findings underscore that the stability of the domestic banking sector plays a crucial role in reducing sovereign risk and its exposure to global factors.

This study contributes to the literature in at least three ways. First, it introduces a new measure of banking fragility in the banking sector (JLoss) that reflects the expected joint loss of the domestic banking sector in the event of a large financial meltdown. The calculation of our JLoss metric employs a saddle-point methodology in which the distribution of potential losses in the banking system is a function of the bank-specific probabilities of default, the exposure in case of default, a loss given default (LGD) parameter, and the correlation between the banks' stock market returns and the stock market index returns of each country. Recent academic studies have introduced measures of systemic risk (e.g., see Brownlees and Engle (2016) for a measure of systemic risk for the U.S.). However, given that our metric of the expected joint loss of the domestic banking sector can be interpreted as the direct cost of bailing banks out from a crisis, it should be a particularly significant factor to consider in the pricing of sovereign bonds.

Second, this study explores the relationship between sovereign credit risk and banking fragility in a sample of emerging economies. Thus, this study is a departure from recent studies that have focused their analysis on samples of European countries during the Eurozone sovereign and banking crises. De Bruyckere et al. (2013) examine contagion between the banking sector and sovereign default risk in Europe from 2007 to 2012, and find that banks with a weak capital buffer, a weak funding structure and less traditional banking activities are particularly vulnerable to risk spillovers. Black et al. (2016) measure the systemic risk of European banks by using a distress insurance premium (DIP), which integrates the characteristics of bank size, probability of default, and correlation. Mody and Sandri (2012) argue that sovereign credit spreads increased in the eurozone with the greater expectation of public support for distressed banks and that this effect was stronger in countries with lower growth prospects and higher debt burdens. Fratzscher and Rieth (2019) show the correlation between CDS spreads of European banks and sovereigns rose from 0.1 in 2007 to 0.8 in 2013, and attribute this higher correlation to a two-way causality between bank credit risk and sovereign credit risk. Although the study of sovereign credit risk in emerging economies has received much attention (Boehmer and Megginson, 1990; Edwards, 1986; Hilscher and Nosbusch, 2010; Longstaff et al., 2011), new research on the relationship between banking fragility and sovereign credit risk in emerging economies has been sparse.

Third, this study takes an additional step beyond the extant literature by exploring a channel (i.e., the fragility of the banking sector) that

amplifies the effect of global (exogenous) factors on sovereign credit risk. Although global financial factors have recently been viewed as push factors in the literature, they have usually been modeled as having homogeneous effects on sovereign credit risk (see, e.g., González-Rozada and Yeyati (2008)). An exception is the study by Georgoutsos and Migiakis (2013), which explores the determinants of euro area sovereign bond yield spreads and find significant country-level heterogeneity on the effects on spreads. Specifically, our analysis suggests that regulations and policies aimed at improving the stability of the domestic banking sector may be helpful in reducing the exposure to global factors, which have become increasingly important in a more financially integrated world.

The study of how the fragility of the banking sector amplifies the effect of global financial factors on sovereign credit spreads in emerging market economies is important. In recent global financial crises, sovereign bond spreads in emerging market economies have been more dramatically affected than the sovereign bond spreads from advanced economies. We argue in this paper that this widening can be partially attributed to the fragility of the banking sector, which led to rapidly growing public debt and balance sheet risks. In view of these stylized facts, we highlight renewed interest on understand the determinants and channels of sovereign bond spreads in emerging economies.

The remainder of the article is organized as follows. Section 2 presents the JLoss methodology utilized to construct our banking fragility measure. Section 3 describes the sample and variables used in this study. Section 4 presents our empirical strategy and reports the main results. Section 5 conducts a set of robustness checks. Finally, Section 6 concludes.

2. Joint Loss (JLoss) measure

To study the relationship between banking fragility and sovereign credit risk, in this work we construct a novel country-level metric of banking fragility (JLoss). JLoss is a model-based semi-parametric metric of the joint loss of the banking sector conditional on a systemic event. Fig. 1 presents an overview of the JLoss methodology.

To calculate our JLoss metric, we first employ a saddle-point methodology that allows us to calculate the aggregated distribution of losses. In this approach the distribution of potential losses in the banking system is a function of the banks' probabilities and exposure at default, a loss given default (LGD) parameter, and the correlation between banks' stock market returns and a systemic component. The individual probabilities of default are calculated following a modification of the Merton's (1974) model. The exposure is proxied by the amount of liabilities of the banks at the moment of default. The LGD for banking debt is set to a 45%, as suggested by the Bank of International Settlements. The key assumption in our approach is that bank risks are uncorrelated, conditional on being correlated with a country-level systemic factor, which in our case is the overall stock market performance of each country. With the distribution of potential losses in the banking system, we calculate each bank's marginal contribution to the total risk. Finally, we normalize these contributions to the total risk with respect to total liabilities.

Appendix A.1 describes the calculation of the banks' default probabilities. Appendix A.2 describes the aggregation of losses with the saddle-point method.

2.1. Parameterization

Table 1 shows the parameterization used in the JLoss calculation, once the individual expected default frequencies are calculated. These parameters define the characteristics of the aggregate distribution of losses and the method implemented. Time span is quarterly, we use

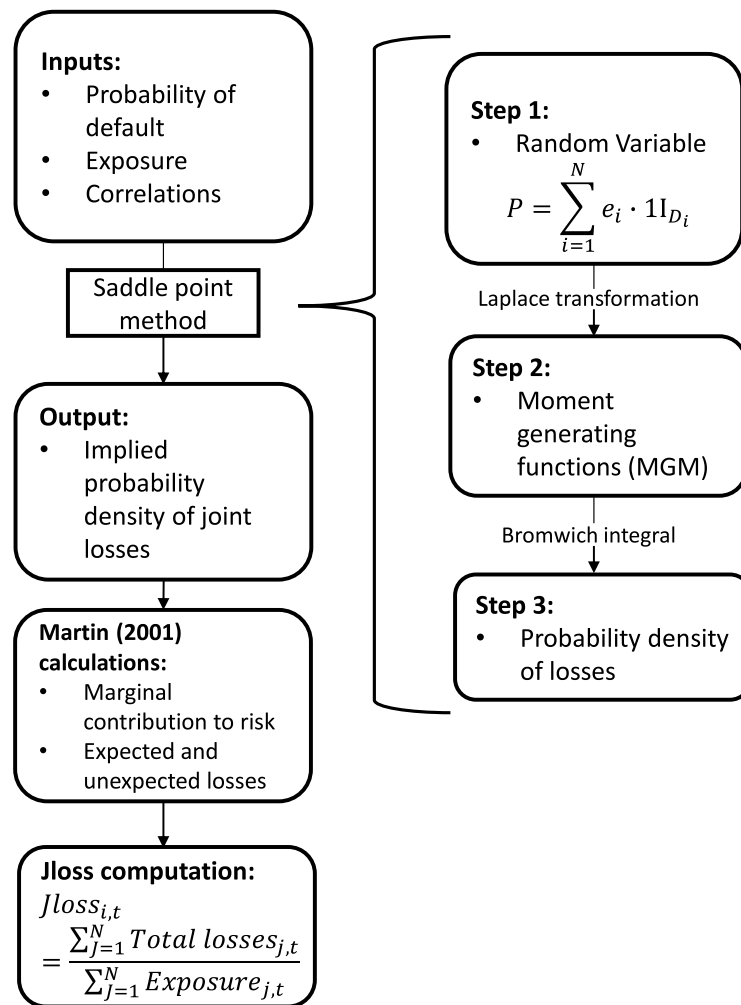


Fig. 1. This figure presents an overview of the JLoss methodology.

Table 1
Parameters saddle-point estimation.

Parameters	Value
Time horizon	1 quarter
Loss given default (LGD)	45%
Approximation nodes	7
Number of systemic factors	1
Losses lower bound	0.010
Losses upper bound	0.048
Number of steps	500
Percentile	0.99
Systemic driver	N(0, 0.16%)

Note: This table reports the parameterization used in the JLoss calculation once the individual expected default frequencies are calculated.

one systematic factor, and seven nodes for the integral quadrature approximation.¹ The lower bound of losses is fixed at 1 percent, whereas the upper bound is assumed to be 4.8 percent. These values are calibrated to losses in the emerging countries' banking systems. Precision

¹ The Gauss–Hermite quadrature solves integrals of the form $I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} f(x) dx$, as the sum $I = \sum_{i=1}^n w_i \cdot f(x_i)$. In our case we are using $n = 7$. Therefore, we need to compute 7 saddle points. In the standard numeric calculus literature, the quadrature is already tabulated to a generic integral. We have just to adjust it to our particular problem.

parameter is the 99 percent. Finally, the systematic factor is assumed to have a normal distribution with zero mean and variations between −4 and 4 percent. Table A.2 in Appendix presents the description and sources of all the variables used in the JLoss computation.

2.2. Discussion

A standard way of calculating the credit risk losses is the methodology described in Vasicek (1997). However, this procedure has some shortcomings that can be improved. The calculations require a functional form of the distribution of losses. This assumption is strong because the estimated parameters of the distribution can lead to important errors in the calculation of losses. Moreover, by being a method that works in the space of real numbers, it lacks of a simple mathematical treatment that allows closed form calculations. The semi-parametric saddle-point approach used in this work, which heavily relies on Martin et al. (2001), has three main advantages. First, it allows simple calculations because it has the ability to provide statistical measures associated directly with credit risk. Second, it significantly increases the speed of calculation in the computational implementation as it can be presented in analytical formulas. Therefore, it allow us to construct our measure for a long number of countries. Third, this method makes it possible to reduce a n-dimensional problem to a single value.

Although Jloss is not the only attempt in the literature to measure financial stability, it is one of the few that performs an aggregation work that allows us to have a metric that reflects financial stability at the country level. For example, the SRISK metric introduced by

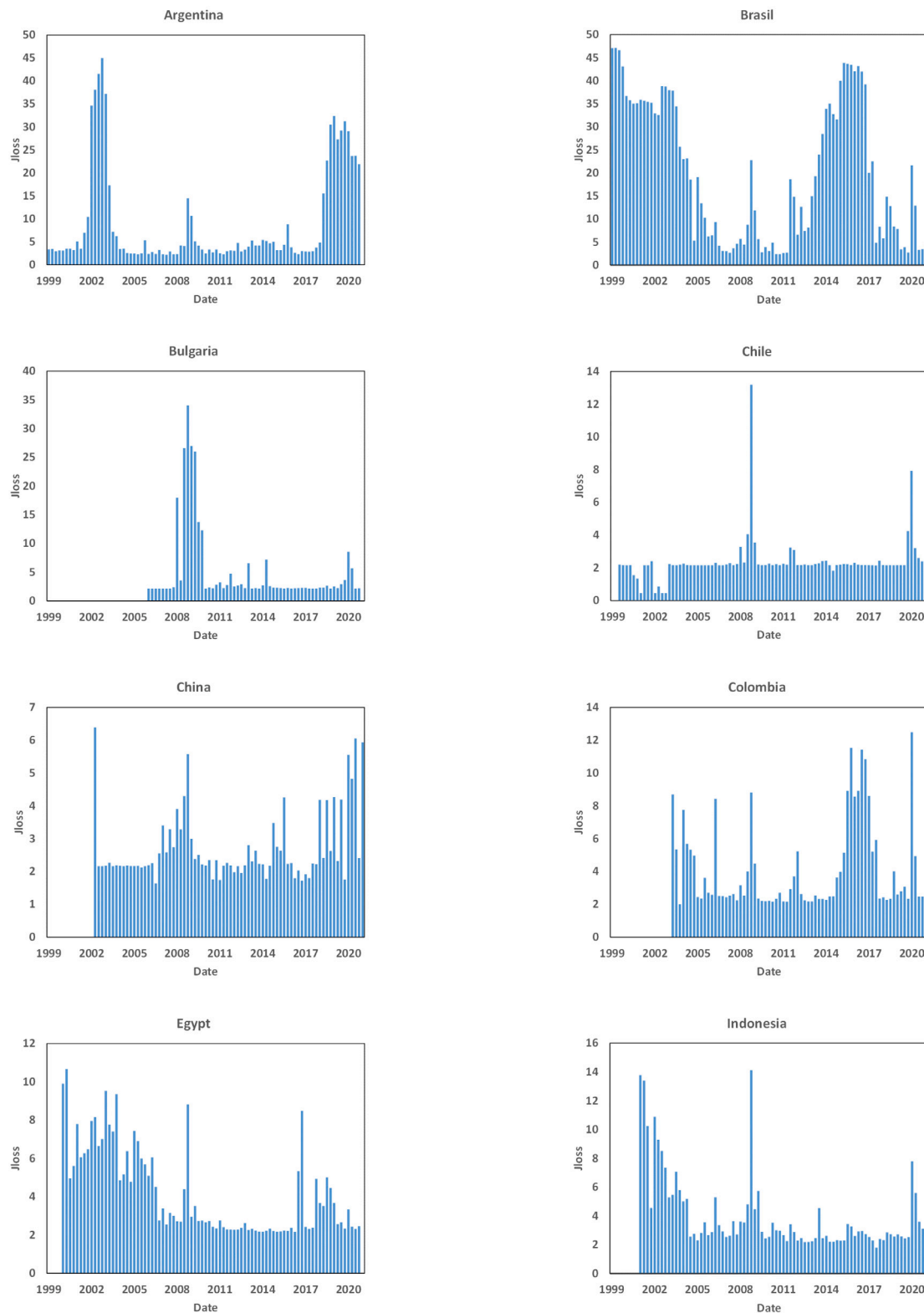


Fig. 2. This figure displays our metric of banking fragility (JLoss) for each of the emerging countries in the sample.

Brownlees and Engle (2016) is an index that computes the expected deficit to the capital of individual financial firms. Brownlees and Engle's (2016) aggregation procedure consists of adding up all the capital losses of a particular financial system. Thus, the aggregate metric does not consider the correlation between the financial institutions. In addition, because the SRISK is a metric based on capital deficits, given

a particular stressed scenario, the metric is more crisis oriented than identifying periods of vulnerability.

The CIMDO-copula introduced by Segoviano and Goodhart (2009) is a metric more similar to the JLoss in methodological terms. However, the difference between the JLoss and the Segoviano CIMDO-copula is that in the first case, the assumptions of conditional independence

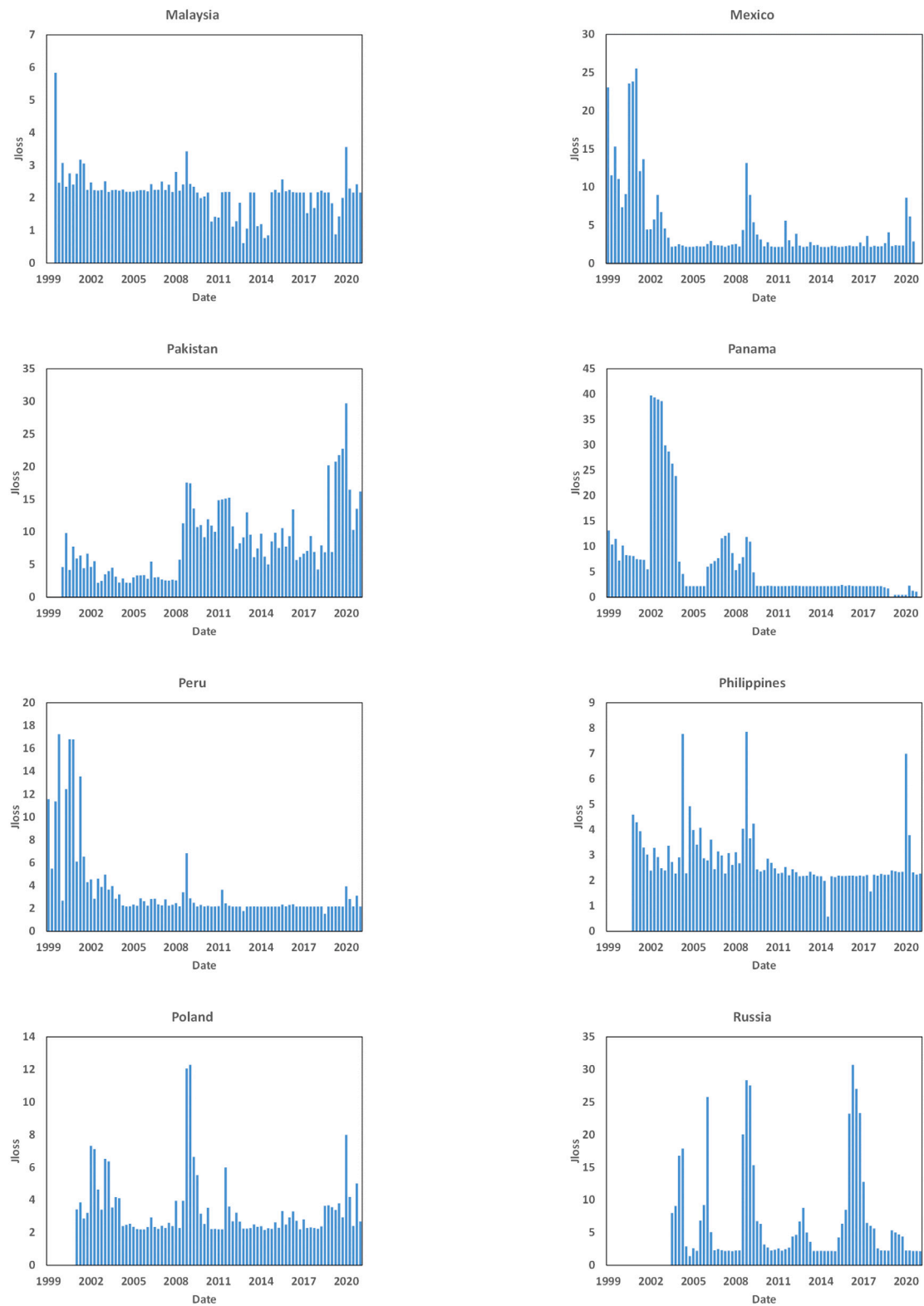


Fig. 2. (continued).

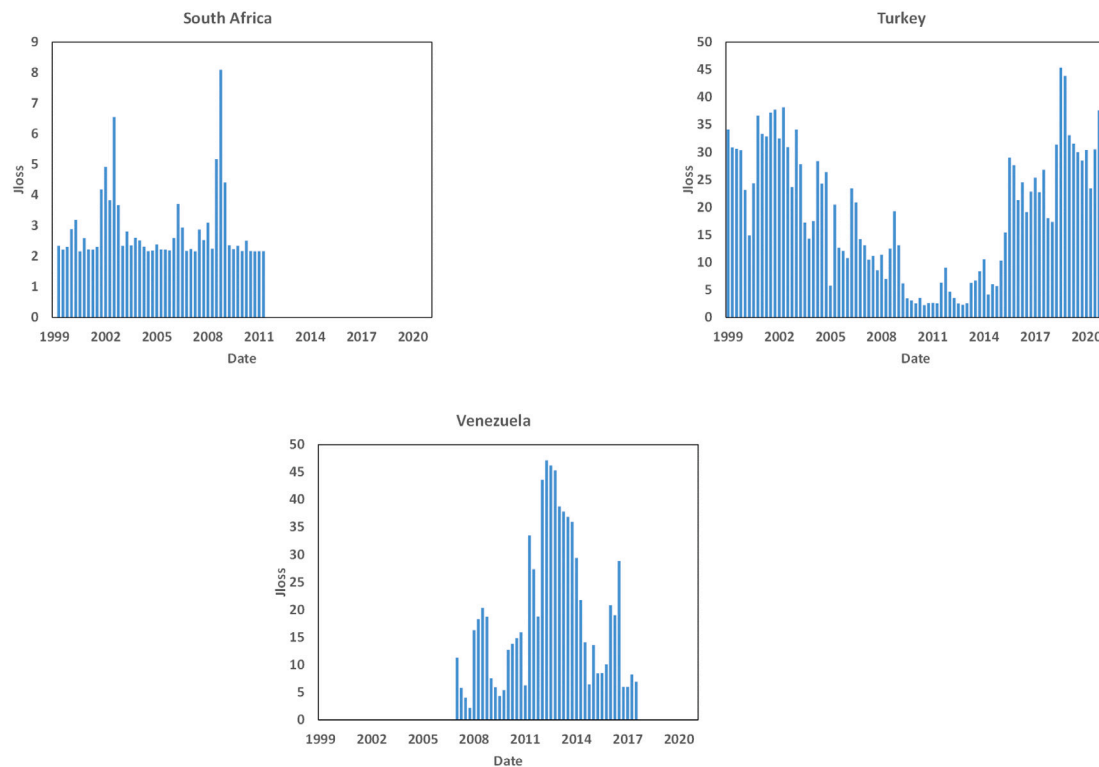


Fig. 2. (continued).

and the semi-parametric calculation allow us to improve efficiency in capturing the changes of variation and offer advantages from the computational point of view, being an approximation but with high precision.

3. Data

To empirically test the relationship between sovereign credit risk, banking fragility, and global factors, we employ a quarterly unbalanced panel database of 19 emerging economies over the period 1999:Q1 to 2021:Q4. The periods covered for each country depend on the availability of all variables used as input to construct our JLoss metric. Our panel data set contains variables related to sovereign credit spreads, financial fragility in the banking sector, country-specific macroeconomic conditions, global financial factors, and measures of systemic risk. The countries in our analysis are those classified as emerging markets in the J. P. Morgan Emerging Markets Bonds Index (EMBI Global) and those for which we had bank-level data to construct the JLoss metric during our sample period. The countries in our sample are: Argentina, Brazil, Bulgaria, Chile, China, Colombia, Egypt, Indonesia, Malaysia, Mexico, Pakistan, Panama, Peru, Philippines, Poland, Russia, South Africa, Turkey, and Venezuela.

Table A.2 in Appendix presents the description and sources of all the variables used in our regression analysis. Our final sample consists of 1406 country-time observations. Table 2 reports summary basic statistics of all the variables used in the regression analysis for the overall sample.

3.1. Sovereign credit risk

The sovereign-credit-risk measure used in this study is the sovereign credit spread. This variable is obtained from the Bloomberg system that collects data from industry sources. Emerging-market sovereign credit spreads are measured using the EMBI Global, which measures the average spread on U.S. dollar-denominated bonds issued by sovereign

entities over U.S. Treasuries. It reflects investors' perception of a government's credit risk. We control for sovereign credit ratings. Our sovereign-credit-rating variable is constructed based on Standard & Poor's (S&P) ratings for long-term debt in a foreign currency.² To compute a quantitative measure of sovereign credit ratings, we follow the existing literature and map the credit-rating categories into 21 numerical values (see, e.g., Borensztein et al. (2013)), with a value of 21 corresponding to the highest rating (AAA) and 1 to the lowest (SD/D).

Table 3 provides summary information for the sovereign credit spreads by country. The average values of the spreads range widely across countries. The lowest average is 135 basis points for China; the highest average is 1405 basis points for Argentina. Both the standard deviations and the minimum/maximum values indicate significant variations also exist over time within countries. For example, the credit spread for Argentina ranges from 204 to 7078 basis points during the sample period.

3.2. Banking fragility

Fig. 2 displays our metric of banking fragility (JLoss) for each of the emerging countries in the sample. As shown in the figure, in most

² Standard and Poor's (2001) defines a foreign-currency credit rating as "A current opinion of an obligor's overall capacity to meet its foreign-currency-denominated financial obligations. It may take the form of either an issuer or an issue credit rating. As in the case of local currency credit ratings, a foreign currency credit opinion on Standard and Poor's global scale is based on the obligor's individual credit characteristics, including the influence of country or economic risk factors. However, unlike local currency ratings, a foreign currency credit rating includes transfer and other risks related to sovereign actions that may directly affect access to the foreign exchange needed for timely servicing of the rated obligation. Transfer and other direct sovereign risks addressed in such ratings include the likelihood of foreign exchange control and the imposition of other restrictions on the repayment of foreign debt".

Table 2
Descriptive statistics.

Variables	N	Mean	Standard deviation	Minimum	Maximum
Sovereign credit risk					
EMBI spread	1406	3.913	6.336	−0.0300	70.78
S&P rating	1406	11.44	3.545	1	18
Banking fragility					
JLoss	1406	6.804	9.057	0	47.16
Control variables					
Profit margin	1348	20.19	36.34	−789.5	138.6
Exchange rate volatility	1329	14.38	61.01	0	932.8
Debt to GDP	1391	45.66	22.21	3.879	147.2
GDP per capita	1406	7021	4278	534	16056
VIX	1406	20.07	8.509	9.510	53.54
U.S. treasury rate	1406	3.211	1.301	0.657	6.442
High yield spread	1406	5.201	2.583	2.380	17.22
On/off-the-run spread	1406	0.114	0.112	−0.0182	0.518

Note: This table reports basic descriptive statistics (observations, mean, standard deviation, minimum and maximum values) of the outcome and control variables used in the regression analysis for the overall sample.

Table 3
Descriptive statistics for sovereign credit spreads.

Country	Mean	Standard deviation	Minimum	Maximum
Argentina	15,60	18,12	2,04	70,78
Brasil	4,36	3,61	1,4	24,12
Chile	1,50	0,54	0,55	3,43
China	1,38	0616	0,44	2,93
Colombia	2,37	0,96	1,12	4,98
Egypt	3,43	1,99	−0,03	8,78
Indonesia	2,51	1,19	0	7,62
Malaysia	1,59	0,63	0,66	3,7
Mexico	2,65	1,16	1,11	6,53
Pakistan	5,96	4,09	0	21,12
Panama	2,54	1,20	1,14	5,61
Peru	2,79	1,79	1,07	8,8
Philippines	2,49	1,58	0,67	6,71
Poland	1,17	0,72	0,12	3,19
Russia	2,43	1,19	0,92	8,05
South Africa	2,17	1,12	0,73	5,62
Turkey	4,06	2,12	1,77	10,66
Venezuela	14,60	8,31	2,07	31,29
Total	3,91	6,33	−0,03	70,78

Note: This table reports descriptive statistics (mean, standard deviation, minimum and maximum values) for the sovereign credit spreads by country.

countries our JLoss metric captures both periods of global financial distress and periods of country-specific idiosyncratic banking fragility. In many countries, idiosyncratic factors seems to have stronger effects on banking fragility than global factors. For example, for the case of Argentina our metric shows that the fragility generated by the sub-prime crisis was smaller than the fragility generated by the 2001 Argentinean sovereign default. In Brazil, idiosyncratic factors such as the “Impeachment” of Dilma Rousseff also seem to have a much stronger effect on banking fragility than global banking fragility.

We believe our model-based semi-parametric measure of banking fragility by country (JLoss) is an important contribution to the literature. Although there are several measures of systemic financial risk available for advanced economies (see Section 3.5 and Giglio et al. (2016)), there is a lack of bank fragility measures for emerging market economies. Our JLoss metric is publicly available.³

³ The JLoss metric for the countries in our sample can be downloaded from <https://www.patriciovalenzuelaaros.com>.

3.3. Country-specific factors

To capture the domestic macro environment, we also control for a set of time-varying country-level macro variables that may directly affect sovereign credit risk: debt to GDP, exchange-rate volatility, profit margin in the banking sector, GDP per capita, and stock market return. As previously mentioned, we also control for the long-term foreign-currency sovereign credit rating. The debt-to-GDP ratio captures the degree of the economy indebtedness. Exchange-rate volatility is the volatility of the country’s exchange rate against the U.S. dollar. We added this variable because it is considered a major determinant of firms’ revenues from abroad and their ability to repay debts denominated in dollars. Profit margin in the banking sector captures the degree of competitiveness in the domestic financial sector. Sovereign credit ratings are credit-rating agencies’ opinion of a government’s overall capacity to meet its foreign-currency-denominated financial obligations. GDP per capita captures the tax base of a country: the greater the tax base of the borrowing country, the greater the ability of a government to repay its debt. The stock market return index of each country is included to control for the pass-through of financial risk to sovereign risk (Gilchrist et al., 2022).

3.4. Global factors

Far from being autarkies, the emerging economies included in this paper have increasingly become more financially integrated with the rest of the world. Therefore, their ability and willingness to serve their debt may depend not only on macroeconomic domestic conditions, but also on the state of the global economy. To capture broad changes in the state of the global (exogenous) financial markets, we consider a set of global financial factors that reflect financial market volatility, risk-free interest rates, risk premiums, and market liquidity. Specifically, the global financial factors used in this study are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. The global factor variables are included in the regressions as the natural logarithm of the variables plus one. For robustness, as we explain in the next section, we also control for a large number of measures of systemic risk proposed in the literature.

The CBOE Volatility Index, known commonly as the VIX, measures the market’s expectation for 30-day volatility in the S&P 500. Usually, a higher VIX indicates either higher physical volatility or higher risk premium and, consequently, an increase in the cost of financing for emerging economies. The 10-year U.S. Treasury rate addresses the interest rate effect. It reflects the risk-free rate against which investors in advanced economies evaluate the payoffs of all other assets of similar maturities. The high-yield spread proxies for the price of risk in the global financial market. We employ J. P. Morgan’s High Yield Spread Index, which measures the yield difference between U.S. dollar-denominated, emerging market high-yield corporate bonds and U.S. Treasuries with similar maturity. Lastly, the On/off-the-run U.S. Treasury spread is the spread between the yield of on-the-run and off-the-run U.S. Treasury bonds. Although the issuer of both types of bonds is the same, on-the-run bonds generally trade at a higher price than similar off-the-run bonds, because of the greater liquidity and specialness of on-the-run bonds in the repo markets.⁴ We compute the On/off-the-run U.S. Treasury spread using 10-year bonds, given that the spread tends to be small and noisy at smaller maturities. The data sources used in the construction of this spread are from Gürkaynak et al. (2007) and the Board of Governors of the Federal Reserve System.

⁴ This specialness arises from the fact that on-the-run Treasury bond holders are frequently able to pledge these bonds as collateral and borrow in the repo market at considerably lower interest rates than those of similar loans collateralized by off-the-run Treasury bonds (Sundaresan and Wang, 2009).

The U.S. variables we employ in this study (i.e., the CBOE Volatility Index, the 10-year U.S. Treasury rate, and the On/off-the-run U.S. Treasury spread) capture global financial factors based on the importance of the U.S. financial system in the global markets and the high commonality between U.S. and global financial variables. Miranda-Agrippino and Rey (2022) argue that U.S. monetary policy shocks induce comovements in global financial variables that characterize the “Global Financial Cycle”.

3.5. Systemic risk measures

For robustness purpose, in some regressions we augment our baseline specification with 19 systemic risk measures: Absorption Ratio, Delta Absorption Ratio, AIM, CoVaR, Delta CoVaR, MES, MES BE, Book Leverage, Market Leverage, CatFin, DCI, Default Yield Spread, International Spillover, GZ Spread, Size Concentration, Realized Volatility, TED Spread, Term Spread, and Turbulence. We also employ the systemic risk index (PQR) of the mentioned systemic risk measures.⁵ Given that these measures are constructed to capture systemic risk stemming from the core of the financial system, these measures are based on data for financial institutions identified by two-digit SIC codes 60 through 67 (finance, insurance, and real estate). Although these measures are available for three regions (US, UK, and EU), for space considerations, we only report the results that use U.S. data. Based on Giglio et al. (2016), the source of these measures, Table A.3 in the Appendix describes each of the systemic risk measures we employ in the robustness checks.

4. Regression analysis and results

The first objective of this study is to explore the relationship between sovereign credit spreads and banking fragility, controlling for other factors that might affect sovereign credit risk independently. We estimate the following baseline econometric model:

$$Spread_{c,t} = \alpha_c + \gamma_t + \beta JLoss_{c,t-1} + \omega X_{c,t} + \epsilon_{c,t}, \quad (1)$$

where $Spread_{c,t}$ is the sovereign credit spread of country c at time t . $JLoss_{c,t-1}$ is the lagged value of our metric of fragility in the banking sector, which computes the joint loss distribution of the banking sector conditional on a systemic event. We employ the lagged value of $JLoss$ to mitigate concerns associated with reverse causality. Both $Spreads_{c,t}$ and $JLoss_{c,t}$ are expressed in natural logarithm. $X_{c,t}$ is a set of time-varying country-level macro variables, including the sovereign credit rating. The term α_c represents a vector of country fixed effects that control for all time-invariant country-specific factors affecting both credit spreads and banking fragility. The term γ_t captures time fixed effects that control for common and global shocks affecting all countries, such as global financial crises or changes in the global financial cycle.⁶ $\epsilon_{c,t}$ is the error term.

Our specification including country fixed effects and time fixed effects is analogous to a difference-in-differences estimator in a multiple-treatment-group and multiple-time-period setting (Imbens and

Wooldridge, 2009).⁷ The identification assumption is that in the absence of domestic banking fragility, the sovereign bond spreads are exposed to similar global shocks. We believe this assumption is plausible, given the homogeneous nature of our sample (i.e., emerging economies that issue international bonds denominated in U.S. dollars) and that global factors are crucial determinants of sovereign credit risk in emerging economies (González-Rozada and Yeyati, 2008).

The second objective of this study is to examine whether the effect of global (exogenous) financial factors on sovereign credit spreads is stronger in countries with more vulnerable banking sectors. To explore this hypothesis, we estimate the following model:

$$Spread_{c,t} = \alpha_c + \beta JLoss_{c,t-1} + \gamma Global_t + \theta JLoss_{c,t-1} \times Global_t + \omega X_{c,t} + \epsilon_{c,t}, \quad (2)$$

where $Global_t$ is a global (exogenous) financial factor at time t . The global factor variables are included in the regressions as the natural logarithm of the variables plus one. The coefficient associated with the interaction term, $JLoss_{c,t-1} \times Global_t$, captures whether the impact of global financial factors on sovereign credit spreads differs in countries with different degrees of banking fragility in their banking sectors. We hypothesize that in a financially integrated world where domestic banks and international capital markets work as substitute sources of capital, a stronger banking sector should attenuate a country's exposure to global financial factors.

4.1. Sovereign bond spreads and banking fragility

Table 4 presents the results from the estimation of Eq. (6) with different sets of control variables. The model is estimated by ordinary least squares (OLS) with robust standard errors. The table also reports the estimates of our econometric model by directly including global financial factors instead of time fixed effects. Time fixed effects control and absorb the effects of any global variable. Therefore, in the specifications with time fixed effect we exclude our global factor variables.

The results suggest sovereign credit spreads are positively related to our metric of banking fragility ($JLoss$). This positive correlation between $JLoss$ and sovereign credit spreads is statistically significant and economically meaningful, even after controlling for country and time fixed effects (column 1), for sovereign credit ratings (column 2), and for the standard determinants of sovereign credit risk (column 3). We also find similar results when we control for our set of global financial factors instead of time fixed effects (column 4) and for the stock market return index of each country (columns 5 and 6). Given that both the spread and the $JLoss$ metric are expressed in natural logarithm, each of our estimated coefficients represent an elasticity. In terms of the economic magnitude of our findings, the coefficient reported in column 1 suggests that, on average, a 100-percent increase in our $JLoss$ measure raises the sovereign credit spread by 22 percent. The coefficient reported in column 3 suggests that a comparable increase in

⁵ According to Giglio et al. (2016), the “PQR condenses the cross section of predictors according to their quantile covariation with the forecast target. By weighting predictors based on their predictive strength, PQR chooses a linear combination that is a consistent quantile forecast”. For more details in the construction of the PQR index, see Giglio et al. (2016).

⁶ Miranda-Agrippino and Rey (2020) show that a single global factor explains a significant share of the common variation of a large cross section of risky asset prices around the world. In our case, the adjusted R-squared of models that consider only time-fixed effects or country- and time-fixed effects are 0.13 and 0.61, respectively. Eq. (6) does not include global factor variables as their effects would be absorbed by the time fixed effects.

⁷ We estimate unit root tests for our sample of sovereign credit spreads once fixed effect are subtracted. For large panels, Choi (2001) proposes the Modified Inverse Chi-square Pm test, which converges to a standard normal distribution. With an average number of ninety-eight periods, our unbalanced panel database is a large panel. Given our baseline specifications control for country fixed effects and time fixed effects, we explore unit roots in the component of sovereign credit spreads that is orthogonal to those fixed effects. That is, the residual from a regression of sovereign credit spreads on country fixed effects and time fixed effects. The Modified Inverse Chi-square Pm test, with panel means, was equal to 2.6577 and its p -value was equal to 0.0039. The Modified Inverse Chi-square Pm test, with panel means and a time trend, was equal to 4.6742 and its p -value was equal to 0.0000. Thus, we reject the null hypothesis of unit root (H_0 : All panels contain unit roots). These results confirm we can estimate our fixed effects models in levels.

Table 4
Sovereign credit spreads and banking fragility.

EMBI spread	(1)	(2)	(3)	(4)	(5)	(6)
JLoss, Lag = 1	0.163*** (0.0248)	0.116*** (0.0193)	0.112*** (0.0203)	0.116*** (0.0200)	0.128*** (0.0206)	0.126*** (0.0201)
Rating S&P		−0.127*** (0.0100)	−0.121*** (0.0104)	−0.126*** (0.0107)	−0.102*** (0.0107)	−0.109*** (0.0113)
Exchange rate volatility			0.0430 (0.0331)	0.0663** (0.0321)	0.0213 (0.0334)	0.0341 (0.0323)
Profit margin			−0.0171 (0.0205)	−0.0480** (0.0201)	0.0215 (0.0213)	−0.0123 (0.0215)
Debt to GDP			0.227*** (0.0506)	0.257*** (0.0489)	0.332*** (0.0552)	0.330*** (0.0537)
GDP per capita			0.131** (0.0619)	0.221*** (0.0438)	0.0376 (0.0614)	0.148*** (0.0446)
Stock market return					−0.370*** (0.112)	−0.513*** (0.0994)
VIX				0.0385 (0.0508)		0.0931* (0.0501)
US treasury				−0.0522 (0.0438)		−0.0597 (0.0438)
US high yield				0.450*** (0.0723)		0.325*** (0.0753)
On/Off-the-run spread				0.0971 (0.168)		0.216 (0.168)
Observations	1036	1036	940	940	907	907
R-squared	0.737	0.818	0.815	0.782	0.784	0.744
Adjusted R-squared	0.714	0.802	0.796	0.776	0.761	0.736
Country FE	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	NO	YES	NO

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss) and a set of control variables. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. The global variables used as control variables are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. Models 1 to 3 control for country and time fixed effects. Model 4 controls for global financial factors instead of time fixed effects. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

JLoss raises the average spread by 13 percent. Our regressions appear to support the view that banking fragility exerts a strong influence on the pricing of emerging-market sovereign bonds.⁸

Despite the high correlation in our country-level control variables, most of the estimated coefficients of our control variables are statistically significant in the expected direction. On the one hand, the results show that sovereign credit ratings and bank profits are negatively related to sovereign credit spreads. On the other hand, the results show exchange rate volatility, indebtedness, economic development, and global premiums are positively related to sovereign credit spreads. Although the positive coefficient associated with GDP per capita is not intuitive, in unreported regressions, we find the coefficient is negative when we do not control for sovereign credit ratings. We conduct this exercise given the very high correlation between sovereign ratings and GDP per capita. Thus, sovereign credit ratings seem to absorb the effect of economic development on sovereign credit spreads.⁹

4.2. Are countries with fragile banking sectors more exposed to global financial shocks?

Although the literature has explored the relevance of external factors as significant determinants of sovereign credit risk in emerging

economies (see, e.g., [González-Rozada and Yeyati \(2008\)](#)), little research has explored the aspects that make a country more or less resilient to sudden changes in the external context. We explore whether global financial factors affect sovereigns differently depending on the fragility of their banking sectors. Given that the emerging economies included in this paper have increasingly become more financially integrated with the rest of the world and that domestic and international capital markets can provide an alternative source of funding that can complement bank financing, we hypothesize that global financial conditions should typically have a smaller effect on countries with more resilient banking sectors.

Table 5 reports the results from the estimation of two specifications of Eq. (7). Columns 1 to 4 report the estimates of our econometric model including time fixed effects. These time fixed effects control for all observable and unobservable global factors. Therefore, these models cannot include global factor variables as their effects are absorbed by the time fixed effects. Columns 5 to 8 consider global financial factors instead of time fixed effects. We consider only one global factor at the time. As before, the model is estimated by ordinary least squares (OLS) with robust standard errors. The positive and statistically significant coefficients associated with the interaction terms in columns 1 to 4 in **Table 5** indicate that a deterioration in global market volatility, risk-free interest rates, high-yield spreads, and aggregate liquidity produce a higher increase in sovereign credit spreads of countries with more fragile banking sectors. These effects are highly statistically significant and economically meaningful. Columns 5 to 8 in the **Table 5**, which consider the direct effects of global financial factors instead of time fixed effects, show almost qualitatively identical results.

To analyze the magnitude of the impact of global factors on sovereign credit spreads across different levels of banking fragility, we show in **Fig. 3** the partial effect of each of our global factors at different levels of banking fragility (i.e., $\gamma + \theta JLoss_{c,t-1}$ from Eq. (7)). These marginal effects are based on the coefficients reported in columns 5 to 8 of **Table 5**. The magnitudes and confidence intervals reported

⁸ **Table A.4** in the **Appendix** reports our baseline models in first differences. Our main results remain qualitatively unchanged. That is, there is a positive relationship between our JLoss metric and sovereign credit spreads. This relationship is statistically significant and economically meaningful.

⁹ Our results of interest are similar if we control for some alternative control variables such as the ratio of primary budget deficits over GDP and FX reserves. Given that the sample size drops dramatically from 940 to 730 observations (i.e., a 22 percent sample size drop) and some countries are excluded from the sample (Russia and Venezuela) when considering these alternative control variables, we have not included them in our baseline specification.

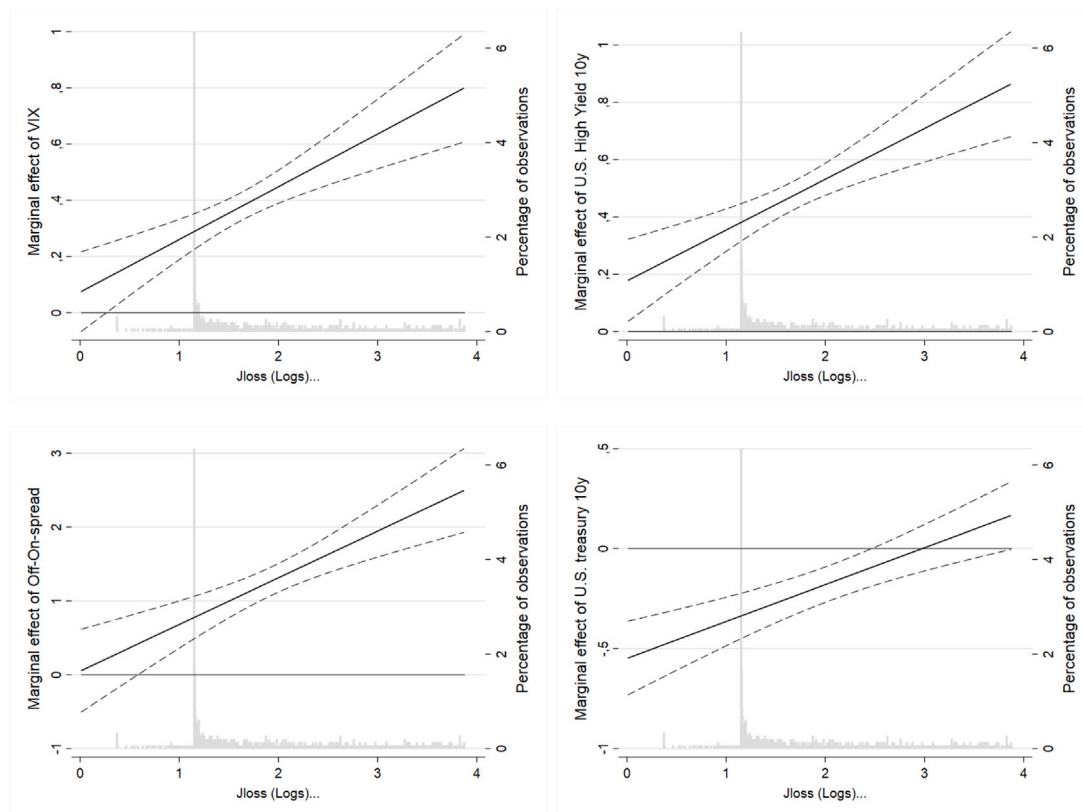


Fig. 3. : Marginal effects of global factors on sovereign credit spreads conditional on values of banking fragility (Jloss). Dotted lines are 95 percent confidence bands. These marginal effects are based on the coefficients reported in columns 5 to 8 of Table 5.

in the figures suggest that the effect of the VIX index, the U.S. high yield, and aggregate illiquidity on sovereign spreads are either zero or positive, and they become positive and larger in economies with more fragile banking sectors. The marginal effects of the U.S. treasury rate on sovereign spreads is negative for low levels of banking fragility and zero for high levels of banking fragility. Given that sovereign spreads measures the average spread on U.S. dollar-denominated bonds issued by sovereign entities over U.S. Treasuries, it is expected the marginal effect of US Treasury rate is negative for low values of banking fragility. Overall, these magnitudes suggest that countries with more fragile banking sectors are more exposed to global (exogenous) financial factors than those with more resilient banking sectors.¹⁰ Thus, our results suggest that stability of the banking sector is a precondition to become more resilient to global shocks and underscore that regulators must ensure the stability of the banking sector to improve governments' borrowing costs in international debt markets.

5. Robustness checks

We conduct five different exercises to check the robustness of our main results. First, we split our sample between periods of financial stability and periods of banking crises. Second, we explore whether our interaction terms may be capturing other non-linear effect of global factors on sovereign credit spreads. Third, we control for a large set of measures of systemic risk proposed in the literature to capture systemic risk stemming from the core of the financial system. Fourth, we control for the average probability of default of the banking system by country.

¹⁰ It is important to note that if the relationship between global factors and sovereign credit spreads is just a simple correlation caused by common macroeconomic factors rather than by a causal effect, global factors should always affect spreads in a similar way.

Finally, we re-estimate our baseline regressions by using sovereign CDS spreads as our dependent variable.

Given that our metric of banking fragility in the banking sector spikes during periods of systemic banking crises, our results are likely driven by a few observations that capture a very high correlation between sovereign risk and banking risk during periods of financial turmoil. Columns 1 of Table 6 reports the results from estimating our baseline regressions for the periods of financial stability. Column 2 reports the results for periods of systemic banking crises. The systemic-banking-crises dummy variable used to divide our sample was constructed using the dataset introduced by Laeven and Valencia (2020).¹¹ The results are qualitatively identical to our baseline regressions reported in Table 4. As expected, the magnitude of our coefficients decrease for the periods of financial stability and increase for the periods of banking crises. However, in both specifications, they remain highly statistically significant and economically meaningful in the expected directions.

Because our primary term of interest in Table 5 is the interaction between JLoss and our four global factors, it is likely that JLoss may captures the effect of another country-specific factor. Table 7 presents the results of a more explicit test of this possibility by including a number of additional interaction terms. The added terms correspond to the interaction of the sovereign credit rating and a banking-crisis dummy variable with our four different measures associated with global factors, respectively. Columns 1 to 4 augment our previous model with the

¹¹ The "Systemic Banking Crises Database II" draws on 151 systemic banking crisis episodes around the globe during 1970–2017. The database, methodology and the periods each country experienced a systemic banking crisis are available in the following link: <https://link.springer.com/article/10.1057/41308-020-00107-3>.

Table 5
Sovereign credit spreads, banking fragility, and global financial factors.

EMBI spread	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Jloss = L,	−0.412*** (0.127)	0.0232 (0.0562)	−0.194*** (0.0743)	0.0474** (0.0232)	−0.422*** (0.125)	−0.106* (0.0584)	−0.197*** (0.0761)	0.0737*** (0.0247)
Rating S&P	−0.121*** (0.0104)	−0.122*** (0.0106)	−0.122*** (0.0104)	−0.123*** (0.0104)	−0.129*** (0.0109)	−0.125*** (0.0112)	−0.127*** (0.0107)	−0.129*** (0.0107)
Exchange rate volatility	0.0516 (0.0325)	0.0459 (0.0333)	0.0514 (0.0327)	0.0597* (0.0331)	0.0740** (0.0333)	0.135*** (0.0355)	0.0812*** (0.0301)	0.133*** (0.0326)
Profit Margin	−0.0256 (0.0199)	−0.0169 (0.0206)	−0.0225 (0.0199)	−0.0190 (0.0199)	−0.0638*** (0.0210)	−0.0750*** (0.0227)	−0.0496** (0.0196)	−0.0574*** (0.0210)
Debt to GDP	0.216*** (0.0517)	0.215*** (0.0514)	0.222*** (0.0509)	0.202*** (0.0511)	0.194*** (0.0516)	0.0802 (0.0528)	0.262*** (0.0486)	0.243*** (0.0514)
GDP per capita	0.127** (0.0622)	0.120* (0.0628)	0.121* (0.0626)	0.0961 (0.0626)	0.239*** (0.0374)	0.0881* (0.0500)	0.248*** (0.0363)	0.366*** (0.0386)
VIX					0.0731 (0.0729)			
US treasury						−0.549*** (0.0945)		
US high yield							0.177** (0.0734)	
On/Off-the-run spread								0.0482 (0.288)
VIX × Jloss (Lag = 1)	0.178*** (0.0431)				0.188*** (0.0421)			
US treasury × Jloss (Lag = 1)		0.0663* (0.0389)				0.184*** (0.0404)		
US high yield × Jloss (Lag = 1)			0.175*** (0.0411)				0.177*** (0.0410)	
On/Off-the-run spread × Jloss (Lag = 1)				0.664*** (0.127)				0.632*** (0.140)
Observations	940	940	940	940	940	940	940	940
R-squared	0.820	0.816	0.819	0.821	0.769	0.720	0.787	0.754
Country FE	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	NO	NO	NO	NO
Adjusted R-squared	0.801	0.796	0.800	0.802	0.763	0.713	0.781	0.747

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss) and a set of control variables. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. The global variables used as control variables are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. Models 1 to 4 control for country and time fixed effects. Models 5 to 8 control for global financial factors instead of time fixed effects. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

Table 6
Periods of financial stability versus periods of crises.

EMBI spread	Periods of financial stability	Periods of banking crises
JLoss, Lag = 1	0.110*** (0.0202)	0.148*** (0.0379)
Rating S&P	−0.121*** (0.0103)	−0.0980*** (0.0193)
Exchange rate volatility	0.0447 (0.0328)	0.0371 (0.0378)
Profit margin	−0.0169 (0.0204)	−0.0187 (0.0248)
Debt to GDP	0.212*** (0.0490)	−0.218 (0.166)
GDP per capita	0.141** (0.0612)	−0.189 (0.208)
Observations	781	159
R-squared	0.817	0.963
Adjusted R-squared	0.798	0.951
Country FE	YES	YES
Time FE	YES	YES

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss) and a set of control variables. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. The models control for country and time fixed effects. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

interaction between global financial factors and sovereign credit ratings. Columns 5 to 8 augment our previous model with the interaction between global factors and banking crises. For space considerations,

we do not report the coefficients associated with the control variables. But, all models control for the full set of control variables considered in our baseline regressions, including sovereign credit ratings. Overall, our main findings remain unchanged. That is, the effects of the VIX index, the High Yield spread and the On/off-the-run spread on sovereign spreads is larger in countries with more fragile banking sectors.

In Table 8, we control for a large collection of systemic risk measures proposed in the literature: Absorption Ratio, Delta Absorption Ratio, AIM, CoVaR, Delta CoVaR, MES, MES BE, Book Leverage, Market Leverage, CatFin, DCI, Default Yield Spread, International Spillover, GZ Spread, Size Concentration, Realized Volatility, TED Spread, Term Spread, and Turbulence. We also employ the systemic risk index (PQR) of the mentioned systemic risk measures. We take these measures from Giglio et al. (2016). As in the previous table, we do not report the coefficients associated with the control variables, but we do control for all the variables.

For space considerations, we only use the measures for the US. The purpose of controlling for alternative measures of global financial distress is to explore the relationship between country-specific idiosyncratic banking fragility and sovereign credit spreads after controlling for global financial distress. Remember that Fig. 2 shows that in most countries our JLoss metric captures both periods of country-specific idiosyncratic banking fragility and periods of global financial distress. The results reported in the table show that the coefficients associated with our JLoss measure remains positive and highly significant in all the 20 models. Our main finding proved to be robust to controlling for a large set of systemic risk measures. Thus, our JLoss metric seems to capture not only systemic risk factors but also idiosyncratic factors that are not captured on global financial measures.

Table 7
Alternative non-linear effects.

EMBI spread	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
JLoss, Lag = 1	−0.595*** (0.160)	0.0846 (0.0730)	−0.291*** (0.0953)	0.0523** (0.0231)	−0.411*** (0.128)	0.0451 (0.0564)	−0.198*** (0.0766)	0.0509** (0.0231)
Banking crisis					3.719 (2.755)	5.831* (3.169)	3.641*** (0.942)	1.709* (0.885)
VIX × JLoss(Lag = 1)	0.240*** (0.0539)				0.177*** (0.0434)			
VIX × Rating S&P	0.0237** (0.0114)							
VIX × Banking crisis					−1.085 (0.826)			
U.S. Treasury rate × JLoss(Lag = 1)		0.0234 (0.0541)				0.0471 (0.0394)		
U.S. Treasury rate × Rating S&P		−0.0184 (0.0171)						
U.S. Treasury rate × Banking crisis						−3.040* (1.681)		
High yield spread × JLoss(Lag = 1)			0.231*** (0.0515)				0.179*** (0.0426)	
High yield spread × Rating S&P			0.0220 (0.0135)					
High yield spread × Banking crisis							−1.652*** (0.441)	
On/off-the-run spread × JLoss(Lag = 1)				0.622*** (0.143)				0.623*** (0.131)
On/off-the-run spread × Rating S&P				−0.0181 (0.0288)				
On/off-the-run spread × Banking crisis								−5.917* (3.206)
Observations	940	940	940	940	940	940	940	940
R-squared	0.821	0.816	0.821	0.821	0.823	0.820	0.825	0.825
Adjusted R-squared	0.802	0.797	0.802	0.802	0.804	0.800	0.807	0.806
Control variables	YES	YES	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES	YES

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss), a comprehensive set of control variables, and interaction terms. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. All regressions control for country and time fixed effects. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

In addition to show that our JLoss metric is statistically significant even after controlling for different measures of systemic risk, [Table 9](#) controls for the country-average probability of default of banks. The employed country-average probability of default of banks is the average of the unconditional probabilities of default derived in Eq. (5). Once we simultaneously control for default probabilities and our JLoss metric, only our JLoss metric remains statistically significant. Although this probability of default of banks is a critical constitute part of the construction of the JLoss measure, this exercise demonstrates that our methodology is advantageous in comparison with a simpler, but more usually employed, measure of banking fragility.

A rich body of research has explored the determinants of both sovereign bond spreads and sovereign CDS spreads. The main advantages of using sovereign CDS spreads are that these spreads only reflect the premium of credit risk and that the sovereign CDS markets tend to be more liquid than the underlying sovereign bond markets ([Longstaff et al., 2011](#)). Thus, CDS spreads are less likely to be influenced, for example, by liquidity premium. Despite the advantages of using sovereign CDS spreads, we prefer to employ EMBI sovereign credit spreads for two reasons. First, the sample size drops considerable when using CDS spreads instead of EMBI sovereign spreads. In fact, the sample size drops dramatically from 1036 to 737 observations (i.e., a 29 percent sample size drop) and some countries (Venezuela and Russia) are excluded from our analysis. Second, our focus is on the cost of financing for governments from emerging economies rather than in the effects in the CDS market. Given that the understanding of the determinants of both sovereign credit spreads and sovereign CDS spreads are important, as an additional robustness check [Table 10](#) reports our baseline

regressions by using a smaller sample of sovereign CDS spreads. The table shows our results are similar when using sovereign bond spreads or sovereign CDS spreads.

6. Conclusion

The global financial crisis of 2008–09 and the European debt crisis generated large losses in the banking sector, triggering a significant deterioration of sovereign credit risk with the greater expectation of public support for distressed banks. These events spurred a renewed interest in generating new measures of banking fragility as well as in understanding the consequences of such vulnerabilities. Despite a new large body of research on the relationship between sovereign risk and bank risk in the eurozone, rigorous research on the nexus between sovereign risk and bank risk in emerging markets is scant. A better understanding of the factors influencing sovereign risk and of how these factors can be properly measured in both advanced and emerging economies is of key importance.

The goal of this paper is to shed light on the relationship between sovereign credit risk and banking fragility in the banking sector. To achieve this goal, we develop a novel model-based semi-parametric metric (JLoss) that computes the joint-loss distribution of a country's banking sector conditional on a systemic event. We find that, controlling for country-level macro variables as well as for country and time fixed effects, our metric of banking fragility (JLoss) is positively associated with sovereign credit spreads and negatively associated with higher sovereign credit ratings in our sample of emerging economies.

Table 8

Systemic risk measures.

EMBI spread	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
JLoss, Lag = 1	0.117*** (0.0200)	0.159*** (0.0256)	0.168*** (0.0255)	0.163*** (0.0261)	0.155*** (0.0255)	0.174*** (0.0259)	0.165*** (0.0254)	0.157*** (0.0253)	0.173*** (0.0262)	0.173*** (0.0255)
CatFin	0.0927 (0.168)									
PQR		3.053 (2.414)								
Absorption Ratio			−1.289*** (0.311)							
AIM				6.653 (5.288)						
Book Leverage					−2.687 (1.908)					
CoVaR						−6.677*** (1.390)				
DCI							−0.945*** (0.309)			
Default Yield Spread								−0.207*** (0.0607)		
Delta Absorption Ratio									1.157*** (0.299)	
Delta CoVaR										−14.13*** (2.650)
Observations	940	510	510	510	510	510	510	510	510	510
R-squared	0.782	0.781	0.790	0.781	0.781	0.786	0.784	0.786	0.789	0.789
Adjusted R-squared	0.776	0.769	0.778	0.769	0.769	0.774	0.771	0.774	0.777	0.777
Control variables	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
EMBI spread	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
JLoss, Lag = 1	0.137*** (0.0289)	0.136*** (0.0292)	0.171*** (0.0256)	0.158*** (0.0254)	0.157*** (0.0255)	0.163*** (0.0258)	0.153*** (0.0261)	0.162*** (0.0261)	0.160*** (0.0256)	0.162*** (0.0259)
GZ Spread	−0.0290 (0.0452)									
Int. Spillover		−0.00200 (0.00218)								
MES			−4.293*** (0.875)							
MES BE				−6.195** (2.844)						
Market Leverage					−0.0217*** (0.00837)					
Realized Volatility						3.983 (2.502)				
Size Concentration							0.169** (0.0750)			
TED Spread								0.000130 (0.000332)		
Term Spread									0.0253* (0.0134)	
Turbulence										0.000248 (0.000412)
Observations	444	427	510	510	510	510	510	510	510	510
R-squared	0.786	0.785	0.787	0.782	0.782	0.781	0.783	0.780	0.781	0.780
Adjusted R-squared	0.772	0.771	0.775	0.770	0.770	0.769	0.770	0.768	0.769	0.768
Control variables	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss) and a comprehensive set of control variables, including 20 alternative measures of systemic risk. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. The global variables used as control variables are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

Table 9
Default probability of the banking sector.

EMBI spread	(1)	(2)	(3)	(4)
Probability of default, Lag = 1	2.526 (1.739)	−0.335 (1.364)	−3.380*** (1.254)	−2.841** (1.272)
Jloss, Lag = 1	0.122*** (0.0334)	0.126*** (0.0264)	0.174*** (0.0289)	0.165*** (0.0287)
Rating S&P		−0.129*** (0.00985)	−0.124*** (0.0104)	−0.129*** (0.0107)
Exchange rate volatility			0.0408 (0.0329)	0.0676** (0.0318)
Profit margin			−0.0124 (0.0201)	−0.0388** (0.0194)
Debt to GDP			0.227*** (0.0474)	0.259*** (0.0452)
GDP per capita			0.0996* (0.0592)	0.214*** (0.0427)
VIX				0.0831* (0.0445)
US treasury				−0.119*** (0.0396)
US high yield				0.347*** (0.0533)
On/Off-the-run spread				0.256* (0.143)
Observations	1084	1084	985	985
R-squared	0.740	0.821	0.820	0.787
Adjusted R-squared	0.718	0.806	0.801	0.780
Country FE	YES	YES	YES	YES
Time FE	YES	YES	YES	NO

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss), the average probability of default in the banking sector, and a comprehensive set of control variables. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. The global variables used as control variables are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

We also explore whether bank stability reduce a country's exposure to global financial factors. A better understanding of the mechanisms through which global factors influence sovereign credit risk is crucial. As highlighted by [González-Rozada and Yeyati \(2008\)](#), emerging economies need to formulate mechanisms to reduce their exposure to global financial factors, as the process of financial integration exhibited over the past four decades brings contagion from other advanced and emerging economies. Our results indicate that countries with more fragile banking sectors are more exposed to the influence of global financial factors.

Our results have important policy implications because they underscore that the stability of a country's domestic banking sector plays a crucial role in reducing sovereign risk and its sensitivity to global factors. Therefore, countries must implement policies oriented to improve the stability of their banking sectors to improve their access to international capital and reduce potentially undesired effects of integration.

This study on the relationship between our metric of banking fragility (JLoss) and sovereign credit spread is a first approach to explore the validity and power of our metric. Thus, our study has some limitations and proposes some issues for future research. An important limitation in our study is the identification strategy. Although we have conducted a comprehensive set of checks to mitigate endogeneity concerns associated with reserve causality and omitted variables, we cannot completely rule out endogeneity biases. Future research should focus on the effect of our JLoss metric on corporate credit risk and firm level performance. More granular data, for example at the firm, should allow researchers to have a cleaner identification strategy to examine causal effects of interest. Our JLoss measure is publicly available to be employed in contexts where it is more feasible to deal with potential endogeneity.

Table 10
Sovereign CDS spreads.

Sovereign CDS spreads	(1)	(2)	(3)	(4)
JLoss, Lag = 1,	0.172*** (0.0308)	0.0938*** (0.0237)	0.107*** (0.0256)	0.102*** (0.0263)
Rating S&P		−0.151*** (0.0101)	−0.151*** (0.0117)	−0.162*** (0.0121)
Exchange rate volatility			−0.0255 (0.0311)	0.0221 (0.0332)
Profit margin			−0.0315 (0.0330)	−0.00687 (0.0327)
Debt to GDP			0.0740* (0.0422)	−0.0422 (0.0485)
GDP per capita			−0.0176 (0.0715)	0.500*** (0.0652)
VIX				−0.0415 (0.0649)
US treasury				−0.0848 (0.0605)
US high yield				0.639*** (0.0916)
On/Off-the-run spread				−0.0265 (0.246)
Observations	737	737	689	689
R-squared	0.987	0.991	0.991	0.988
Adjusted R-squared	0.985	0.990	0.990	0.987
Country FE	YES	YES	YES	YES
Time FE	YES	YES	YES	NO

Note: This table reports estimates from panel regressions of sovereign CDS spreads against banking fragility (JLoss), and a comprehensive set of control variables. The macro variables used as controls are the sovereign rating, the debt to GDP ratio, the exchange-rate volatility, the profit margin in the banking sector, and the GDP per capita. The global variables used as control variables are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

Another limitation of our study is associated with the understanding of more specific channels through which global factors influence sovereign credit spreads. Our study takes an additional step beyond the extant literature by exploring whether the fragility of the banking sector amplifies the effect of global financial factors on sovereign credit risk. But it would be interesting, for example, to examine whether emerging economies finance their budget deficits through privileged access to their banking sector. Given the study of more detailed channels certainly deserves further inquiry, we believe they can be considered as a promising area for future research.

CRedit authorship contribution statement

Anusha Chari: Investigation, Methodology, Writing – original draft, Writing – review & editing. **Felipe Garcés:** Investigation, Methodology, Writing – original draft. **Juan Francisco Martínez:** Conceptualization, Investigation, Methodology, Writing – original draft. **Patricio Valenzuela:** Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Data availability

Data will be made available on request.

Appendix

A.1. Individual probabilities: distance-to-default

To calculate default probabilities for each bank, we employ [Kealhofer and Kurbat's \(2000\)](#) approach. This approach is a standard modification of the structural credit risk model introduced by [Merton](#)

Table A.1
Banks per country.

Country	Number of banks
Argentina	5
Brazil	11
Bulgaria	3
Chile	6
China	13
Colombia	6
Egypt	9
Indonesia	23
Malaysia	7
Mexico	4
Pakistan	20
Panama	2
Peru	5
Philippines	10
Poland	10
Russia	5
South Africa	3
Turkey	8
Venezuela	3

Note: This table reports the number of banks by country employed to construct the JLoss metric.

(1974). Table A.1 in the appendix reports the number of banks used in our analyses by country.

Our measurement approach merges together information on banks' balance sheet and market prices: long and short term liabilities (L_{ST} , L_{LT}), short term assets (A_{ST}), average interest rates (r), time horizon (T), volatility of bank realized returns (σ_V), and market capitalization (E). With this data we construct the default point (D^*), which we formally define as¹²:

$$D^* = L_{ST} + \frac{1}{2} L_{LT}.$$

Then we numerically solve the following system of two non-linear equations, by using the Newton–Raphson algorithm (Press et al., 2007), to project banks' value of assets ($\hat{V}$) and implied asset volatility ($\hat{\sigma}_A$):

$$\frac{V}{E} \Phi(d_1) - \frac{e^{-rT} \Phi(d_2)}{E/D^*} - 1 = 0$$

$$\Phi(d_1) \frac{V}{E} \sigma_A - \sigma_E = 0.$$

where $d_1 = \log\left(V \frac{E}{D^*}\right) + \frac{\frac{1}{2}\sigma_E^2 T}{\sigma_E \sqrt{T}}$ and $d_2 = d_1 - \sigma_E \sqrt{T}$. Φ stands for the cumulative normal distribution function.¹³

Once we get the projected values $\hat{V}$ and $\hat{\sigma}_A$, we insert them into the following distance to default DD equation:

$$DD = \frac{\frac{\hat{V}}{E} - D^*}{\frac{\hat{V}}{E} \hat{\sigma}_A}.$$

This equation is a function of the predicted value of the banks' assets ($\hat{V}$) and asset volatility ($\hat{\sigma}_A$). Finally, we assume normality to obtain the expected default frequency (EDF) as

$$EDF = \Phi(-DD).$$

We compute this quantity for all banks in every country and time periods of our sample, and associate the expected default frequency

value to the *unconditional* probability of default (p_{def_i}), which is one of the inputs for the saddle-point method.

A.2. Saddle-point method and implementation

The saddle-point method allows us to simplify the calculations of the aggregate distribution of losses by working in a different space. We move from the real numbers space ($\mathbb{R}$) to the moment generating function space (MGF). Then, we apply a transform to come back to the real numbers space. The saddle-point method allows to calculate the distribution of a random variable P that represents the aggregate losses for a portfolio of N banks. Formally, we define P as

$$P = \sum_{i=1}^N e_i \mathbb{1}_{D_i},$$

where e_i is the exposure of bank i , and $\mathbb{1}_{D_i}$ is the indicator function that takes a value of zero if banks have repayment capacity and it is equal to one otherwise.

We need a workable description of the problem in the space of a MGF. To determine the MGF, we assume a feasible functional form that is statistically equivalent to the problem in the real and one-dimensional space ($\mathbb{R}$). The Laplace transform naturally connects the two spaces (from $\mathbb{R}$ to MGF), while that the *Bromwich integral* does the reverse process (from MGF to $\mathbb{R}$). This regularity provides a computational advantage with respect to other methods as it allow us to reduce the dimensionality of the problem.¹⁴

For an arbitrary credit portfolio, the relationship between the probability density functions and the MGF is described as

$$M_x(s) = \mathbb{E}(e^{sx}) = \int e^{sx} f(x) dx,$$

where M_x is the expected value of exponential function (e^{sx}), x is the random variable (of losses, analogous to P), s is the arbitrary Laplace transform parameter, and f represents the probability density function.

If we consider two states for the random variable x (default and no default), we have the following discrete MGF:

$$M_i(s) = \mathbb{E}(e^{si}) = \sum_{\mathbb{1}_{D_i}=0,1} f(\mathbb{1}_{D_i}) e^{s \cdot \text{expos}_i \cdot \mathbb{1}_{D_i}} = 1 - p_{def_i} + p_{def_i} e^{s \cdot \text{expos}_i},$$

where p_{def_i} is the *unconditional* default probability and expos_i is the exposure in the defined time horizon for bank i . If we assume *conditional independence*, the relationship between the *unconditional* (p_{def_i}) and *conditional* ($p_{def_i}(\vec{V})$) probabilities of default can be expressed as¹⁵

$$p_{def_i} = \sum_k p_{def_i}(\vec{V}_k) h(\vec{V}_k), \quad (3)$$

where $\vec{V}_k$ represents the k th set of values of the underlying group of M systemic factors, $\vec{V} = \{V^1, V^2, \dots, V^M\}$. Moreover, $h(\vec{V})$ are the probability density of the systemic factor. Following Koyloughlu and Hickman (1998), we can write $h(\vec{V}) = h^1(V^1) \cdot h^2(V^2) \dots h^M(V^M)$ as the systemic factors are assumed to be uncorrelated. In this work we consider only one country-level systemic factor: the stock market index return of each specific country.

Without loss of generality and consistent with our method of estimation for the individual probabilities of default, we consider a unifactorial Merton-style model.¹⁶ As in Vasicek (1997), we assume that

¹² We follow the KMV approach to choose the value of 0.5 in front of long-term debt. According to the KMV approach, this value empirically fits well the data on corporate defaults.

¹³ We use the realized variance approach to estimate the quarterly equity volatility. Following Barndorff-Nielsen et al. (2002), we compute square root of the sum of squared daily equity returns over a quarter. That is, for every quarter and bank, we calculate $\sigma_E = \sqrt{\sum_{t=1}^Q r_t^2}$, where Q is the number of days in a particular quarter.

¹⁴ Similarly to Martin et al. (2001), when we calculate the *Bromwich integral* through the saddle-point we are taking only the real part of the results since the original results could have imaginary factors.

¹⁵ Conditional independence means that conditional on being correlated to a systemic factor, the banks have uncorrelated probabilities of default. We acknowledge a potential complexity if systemic factors are correlated. However, we assume that they are calculated as orthogonal factor loadings.

¹⁶ This method can be easily extended to allow for multi-factor models.

Table A.2
Description of variables.

Name	Level	Description	Frequency	Source
Regression analysis				
EMBI spread	Country	J.P. Morgan EMBI Global spread (in log)	Quarterly	Bloomberg
S&P rating	Country	S&P sovereign credit rating, long-term debt, foreign currency, 21 = AAA - 1 = SD (in log)	Quarterly	Bloomberg
GDP per capita	Country	USD GDP per capita (in log)	Quarterly	IFS
Debt to GDP	Country	Debt divided by GDP (in log)	Quarterly	IFS
Profit margin	Country	Profit margin (in log)	Quarterly	Bloomberg
Exchange rate volatility	Country	Exchange rate volatility (percentage points)	Quarterly	Bloomberg
Probability of default	Country	Average probability of default of banks	JLoss methodology	
VIX	Global	CBOE Volatility Index (in log)	Quarterly	Bloomberg
U.S. Treasury rate	Global	U.S Treasury yield 10 years (in log)	Quarterly	Bloomberg
High yield spread	Global	J.P. Morgan high yield spread (in log)	Quarterly	Bloomberg
On/off-the-run spread	Global	Difference between the yield to maturity of 10 years off-the-run and on-the-run Treasury bonds (in log) the yields from a smooth zero-coupon yield curve (in log)	Quarterly	Board of Governors of the Federal Reserve System
JLoss computation				
Stock market index	Country	Stock market index	Daily	Bloomberg
Stock price returns	Bank	Stock price returns	Daily	Bloomberg
Long term liabilities	Bank	Long term liabilities	Quarterly	Bloomberg
Short term liabilities	Bank	Short term liabilities	Quarterly	Bloomberg
Average bank interest rates	Bank	Average bank interest rates	Quarterly	Bloomberg
Market capitalization	Bank	Market capitalization	Quarterly	Bloomberg
Volatility of stock price	Bank	Volatility of stock price	Quarterly	Bloomberg
Correlation to systemic factor	Bank	Correlation stock return	Quarterly	Bloomberg

Table A.3
Systemic risk measures.

Variables	Source	Description
CoVaR and Delta CoVaR	Adrian and Brunnermeier (2011)	CoVaR is defined as the value-at-risk (VaR) of the financial system as a whole conditional on an institution being in distress. Delta CoVaR is defined as the VaR of the financial system when institution i is at quantile q (in distress) relative to the VaR when institution i is at the median of its distribution.
MES	Acharya et al. (2010)	These measures capture the exposure of each individual firm to shocks to the aggregate system. MES captures the expected return of a firm conditional on the system being in its lower tail.
MES BE	Brownlees and Engle (2011)	This version of MES employs dynamic volatility models to estimate the components of MES.
CatFin	Allen et al. (2012)	This measure computes the time varying VaR of financial institutions at the 99 percent confidence level, using the cross-sectional distribution of returns on the equity of financial firms in each period.
Absorption Ratio	Kritzman et al. (2010)	This measure computes the fraction of return variance of a set of N financial institutions explained by the first K < N principal components.
Delta Absorption Ratio	Kritzman et al. (2010)	It is the difference between absorption ratios calculated for long and short estimation windows.
DCI	Billio et al. (2012)	The dynamic causality index aims to capture how interconnected a set of financial institutions is by computing the fraction of significant Granger-causality relationships among their returns.
International Spillover	Diebold and Yilmaz (2009)	The index aggregates the contribution of each variable to the forecast error variance of other variables across multiple return series. It captures the total extent of spillover across the series considered (a measure of interdependence).
Book and Market Leverage	Giglio et al. (2016)	They construct a measure of aggregate book leverage (debt/assets) and aggregate market leverage (debt/market equity) for the largest 20 financial institutions to capture the potential for instability and shock propagation that occurs when large intermediaries are highly levered.
Size Concentration	Giglio et al. (2016)	They construct the Herfindal index of the size distribution among financial firms.
Turbulence	Kritzman and Y. Li (2010)	Turbulence is a measure of “excess volatility”, that compares in each period the realized squared returns of financial institutions with their historical volatility.
AIM TED Spread	Amihud (2002) Giglio et al. (2016)	AIM captures a weighted average of stock-level illiquidity. The difference between three-month LIBOR and three-month T-bill interest rates.
Default Yield Spread	Giglio et al. (2016)	The difference between yields on BAA corporate bonds and Treasuries.
GZ Spread	Gilchrist and Zakrajsek (2012)	It is an alternative measure of credit spread constructed from individual unsecured corporate bonds, where the yield of each bond is compared to the yield of a synthetic treasury bond with the same cash flows to obtain a precise measure of its credit spread. The individual credit spreads are then averaged across all maturities and all firms.
Term Spread	Global Financial Data	The difference between yields on ten year and one month US Treasury bonds.
Realized Volatility	Giglio et al. (2016)	It is the aggregated series of realized volatility by averaging the individual volatility (within month standard deviation of daily returns) across the 20 largest institutions.
Insolvency	Giglio et al. (2016)	It is constructed by aggregating series of insolvency (i.e., the inverse of the within-month standard deviation of daily returns) by averaging them across the 20 largest institutions.

Note: This table describes the systemic risk measures used in the robustness checks. The source is [Giglio et al. \(2016\)](#).

Table A.4
Sovereign credit spreads and banking fragility.

Δ EMBI	(1)	(2)	(3)	(4)
Δ Jloss	0.0415*** (0.0156)	0.0320** (0.0131)	0.0235* (0.0137)	0.0412*** (0.0133)
Δ Rating S&P		-0.135*** (0.0463)	-0.0583** (0.0246)	-0.0547** (0.0258)
Δ Exchange rate			0.0485** (0.0211)	0.0644*** (0.0223)
Δ Profit margin			-0.0201** (0.0102)	-0.0252** (0.0100)
Δ VIX				0.0615** (0.0301)
Δ US treasury				0.0607 (0.0579)
Δ US high yield				0.511*** (0.0518)
Δ On/off-the-run spread				0.286*** (0.106)
Observations	1031	1031	920	920
R-squared	0.520	0.594	0.602	0.491
Adjusted R-squared	0.477	0.557	0.561	0.477

Note: This table reports estimates from panel regressions of sovereign credit spreads against banking fragility (JLoss) and a set of control variables. The macro variables used as controls are the sovereign rating, the exchange-rate volatility, and the profit margin in the banking sector. The global variables used as control variables are the CBOE Volatility Index, the 10-year U.S. Treasury rate, the 10-year U.S. High Yield spread, and the On/off-the-run U.S. Treasury spread. Robust standard errors are in parentheses below each coefficient estimate. ***, **, and * indicate significance at the 1, 5, and 10 percent levels, respectively.

$h(\vec{V})$ follows a Normal distribution. Although stock index returns have been shown to be negatively skewed and leptokurtic, the Normal distribution assumption allow us to simplify the analytical tractability of our model. Thus, the conditional probability in Eq. (3) can be written as:

$$p_{def_i}(V) = P\left(Z \leq \Phi^{-1}\left(p_{def_i}(V)\right)\right) = \Phi\left(\frac{\Phi^{-1}\left(p_{def_i}\right) - \rho V}{\sqrt{1 - \rho^2}}\right),$$

where ρ is the correlation between the individual banks' stock market returns and the stock market index return of each specific country. After these calculations, we are able to define the conditional and unconditional MGF, as a function of the underlying country-level systemic factor:

$$M(s|V) = \prod_{i=1}^N M_i(s) = \prod_{i=1}^N \left(1 - p_{def_i}(V) \left(e^{\text{expos}_i s}\right)\right). \quad (4)$$

In order to further simplify the calculations, we use the *cumulant* generating functions (K), defined as the logarithm of the MGF. Thus, $K(s|V) = \log(M(s|V))$. The useful property of this function is that all moments of the distribution described by the probability density $f(\cdot)$ can be generated by calculating the derivatives evaluated at $s = 0$. For instance, for the two first moments we have $K'(s = 0) = \mathbb{E}(x)$ and $K''(s = 0) = \mathbb{V}\text{ar}(x)$.

With the correlation matrix of the stock returns of the individual banks and the market, we are able to obtain the MGF in Eq. (4). Next, we reverse the process to come back to the space of real numbers and get the joint probability density of losses. To do that we employ the *Bromwich* integral. Under our *conditional independence* assumption, this integral takes the form:

$$f(x) = \frac{1}{2\pi i} \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} e^{K(s|V) - sx} ds \right) h(V) dV.$$

To solve the above integral, we use a particular property. Close to the saddle-point of the argument of the exponential function, the integral can be approximated with high level of accuracy. If we obtain the first order conditions for the argument of the exponential, we obtain

that $\frac{d}{ds}(K(s) - sx)$, and $K'(s = \hat{r}_V) = x$. In the previous expression, $\hat{r}$ is the saddle point of the integral.

The expression in Eq. (3) in the continuous case becomes:

$$P(L > x) = \int_{-\infty}^{+\infty} P(L > x|V) h(V) dV \\ = \frac{1}{2\pi i} \int_{-\infty}^{+\infty} \left(\int_{-\infty}^{+\infty} e^{K(s|V) - sx} ds \right) h(V) dV. \quad (5)$$

With the use of the saddle-point property, the distribution of portfolio losses can be approximated by:

$$P(L > x) \approx \begin{cases} e^{(K(\hat{r}_V|V) - x\hat{r}_V + \frac{1}{2}\hat{r}_V^2 K''(\hat{r}_V))} \Phi\left(-\sqrt{\hat{r}_V^2 K''(\hat{r}_V)}\right), & \text{if } x \leq \mathbb{E}(L) \\ \frac{1}{2}, & \text{if } x = \mathbb{E}(L) \\ 1 - e^{(K(\hat{r}_V|V) - x\hat{r}_V + \frac{1}{2}\hat{r}_V^2 K''(\hat{r}_V))} \Phi\left(-\sqrt{\hat{r}_V^2 K''(\hat{r}_V)}\right), & \text{if } x > \mathbb{E}(L). \end{cases}$$

In order to be able to manage the integral approximation, we need to discretize the expression in (5). For the general case, in a multi-factor setting, we would have:

$$P(L > x) \approx \sum_{k_1} \dots \sum_{k_M} P(L > x|\vec{V} = \{V_{k_1}, \dots, V_{k_M}\}) h(V_{k_1}) \dots h(V_{k_M}) \quad (6)$$

Recall that in our case M , the number of country-level systemic factors, is set to one. We solve the expression in (6) by using a Gauss-Hermite quadrature. By applying the Bayes theorem in (6), we get:

$$P(L > x) \approx \sum_j P(j) P(L > x|j) h(V_{k_1}) \dots h(V_{k_M}), \quad (7)$$

where j is the state of the underlying country-level systemic factor, thus $P(L > x|j)$ is the probability that the losses are greater than x for the systemic factor configuration V . $P(j)$ is the probability that the economy latent variable V is in the state j and it corresponds to the quadrature weight h_{k_j} . The marginal contributions to the overall risk, from a particular bank to the entire financial system of a country, are obtained following Martin (2001). Finally, to obtain our JLoss metric, we normalized with respect to the total liabilities.

References

- Acharya, V., Hickman, A., Schnabl, P., 2014. A pyrrhic victory? Bank bailouts and sovereign credit risk. *J. Finance* 69 (6), 2689–2739.
- Acharya, V., Pedersen, L., Philippon, T., Richardson, M., 2010. Measuring systemic risk. Working Paper, NYU.
- Adrian, T., Brunnermeier, M., 2011. Covar. Working Paper, National Bureau of Economic Research.
- Allen, L., Bali, T., Y., T., 2012. Does systemic risk in the financial sector predict future economic downturns. *Rev. Financ. Stud.* 25 (10), 3000–3036.
- Amihud, Y., 2002. Illiquidity and stock returns: cross-section and time-series effects. *J. Financ. Mark.* 5 (1), 31–56.
- Ang, A., Longstaff, F.A., 2013. Systemic sovereign credit risk: Lessons from the US and Europe. *J. Monet. Econ.* 60 (5), 493–510.
- Barndorff-Nielsen, O., Nicolato, E., Shepard, N., 2002. Some recent developments in stochastic volatility modelling. *J. Quant. Financ.* 2 (1), 11–23.
- Billio, M., Lo, A., Getmansky, M., Pelizzon, L., 2012. Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *J. Financ. Econ.* 104, 535–559.
- Black, L., Correa, R., Huang, X., Zhou, H., 2016. The systemic risk of European banks during the financial and sovereign debt crises. *J. Bank. Financ.* 63, 107–125.
- Boehmer, E., Megginson, W.L., 1990. Determinants of secondary market prices for developing country syndicated loans. *J. Finance* 45 (5), 1517–1540.
- Borensztein, E., Cowan, K., Valenzuela, P., 2013. Sovereign ceilings “Lite”? The impact of sovereign ratings on corporate ratings. *J. Bank. Financ.* 37 (11), 4014–4024.
- Brownlees, C., Engle, R., 2011. Volatility, correlation and tails for systemic risk measurement. Working Paper, NYU.
- Brownlees, C., Engle, R.F., 2016. SRISK: A conditional capital shortfall measure of systemic risk. *Rev. Financ. Stud.* 30 (1), 48–79.
- Cavallo, E., Valenzuela, P., 2010. The determinants of corporate risk in emerging markets: an option-adjusted spread analysis. *Int. J. Financ. Econ.* 15 (1), 59–74.
- Choi, I., 2001. Unit roots tests for panel data. *J. Int. Money Finance* 20 (2), 249–272.

- De Bruyckere, V., Gerhardt, M., Schepens, G., Vander Vennet, R., 2013. Bank/sovereign risk spillovers in the European debt crisis. *J. Bank. Financ.* 37 (12), 4793–4809.
- Diebold, F.X., Yilmaz, K., 2009. Measuring financial asset return and volatility spillovers, with application to global equity markets. *Econ. J.* 119, 158–171.
- Dieckmann, S., Plank, T., 2012. Default risk of advanced economies: An empirical analysis of credit default swaps during the financial crisis. *Rev. Financ.* 16 (4), 903–934.
- Edwards, S., 1986. The pricing of bonds and bank loans in international markets: An empirical analysis of developing countries' foreign borrowing. *Eur. Econ. Rev.* 30 (3), 565–589.
- Farhi, E., Tirole, J., 2012. Collective moral hazard, maturity mismatch, and systemic bailouts. *Amer. Econ. Rev.* 102 (1), 60–93.
- Fratzsch, M., Rieth, M.H., 2019. Monetary policy, bank bailouts and the Sovereign-Bank risk nexus in the Euro Area. *Rev. Financ.* (23), 745–775.
- Georgoutsos, D., Migiakis, P., 2013. Heterogeneity of the determinants of euro-area sovereign bond spreads; what does it tell us about financial stability? *J. Bank. Financ.* 37 (11), 4650–4664.
- Giglio, S., Kelly, B., Pruitt, S., 2016. Systemic risk and the macroeconomy: An empirical evaluation. *J. Financ. Econ.* 119 (3), 457–471.
- Gilchrist, S., Wei, B., Yue, V., Zakrajšek, E., 2022. Sovereign risk and financial risk. *J. Int. Econ.* 136 (103603).
- Gilchrist, S., Zakrajšek, E., 2012. Credit spreads and business cycle fluctuations. *Am. Econ. Rev.* 102 (4), 1692–1720.
- González-Rozada, M., Yeyati, E.L., 2008. Global factors and emerging market spreads. *Econ. J.* 118 (533), 1917–1936.
- Gürkaynak, R.S., Sack, B., Wright, J.H., 2007. The US treasury yield curve: 1961 to the present. *J. Monetary Econ.* 54 (8), 2291–2304.
- Hilscher, J., Nosbusch, Y., 2010. Determinants of sovereign risk: Macroeconomic fundamentals and the pricing of sovereign debt. *Rev. Financ.* 14 (2), 235–262.
- Imbens, G.W., Wooldridge, J.M., 2009. Recent developments in the econometrics of program evaluation. *J. Econ. Lit.* 47 (1), 5–86.
- Kallestrup, R., Lando, D., Murgoci, A., 2016. Financial sector linkages and the dynamics of bank and sovereign credit spreads. *J. Empir. Financ.* 38, 374–393.
- Kealhofer, S., Kurbat, M., 2000. Benchmarking Quantitative Default Risk Models: A Validation Methodology. Research Paper, Moody's KMV.
- Koylougou, H., Hickman, A., 1998. Reconcilable differences. *Risk* 11 (10), 56–62.
- Kritzman, M., Li, Y., Page, S., Rigobon, R., 2010. Principal components as a measure of systemic risk. *J. Portf. Manag.* 37, 112–126.
- Kritzman, M., Y. Li, Y., 2010. Skulls, financial turbulence, and risk management. *Financ. Anal. J.* 66 (5), 30–41.
- Laeven, L., Valencia, F., 2020. Systemic banking crises database II. *IMF Econ. Rev.* 68, 307–361.
- Longstaff, F.A., Pan, J., Pedersen, L.H., Singleton, K.J., 2011. How sovereign is sovereign credit risk? *Am. Econ. J.: Macroecon.* 3 (2), 75–103.
- Martin, R., Thompson, K., Browne, C., 2001. Taking to the saddle: An analytical technique to construct the loss distribution of correlated events. *Risk-Lond.-Risk Mag. Ltd.* 14 (6), 91–94.
- Merton, R.C., 1974. On the pricing of corporate debt: The risk structure of interest rates. *J. Finance* 29 (2), 449–470.
- Miranda-Agrippino, S., Rey, H., 2022. The global financial cycle. In: Gopinath, G., Helpman, E., Rogoff, K. (Eds.), *Handbook of International Economics: International Macroeconomics*, Volume 6. In: *Handbook of International Economics*, 6, Elsevier, pp. 1–43.
- Mody, A., Sandri, D., 2012. The eurozone crisis: How banks and sovereigns came to be joined at the hip. *Econ. Policy* 27 (70), 199–230.
- Press, W.H., Teukolsky, S.A., Vetterling, W.T., Flannery, B.P., 2007. *Numerical Recipes 3rd Edition: The Art of Scientific Computing*. Cambridge University Press.
- Segoviano, M., Goodhart, C., 2009. Banking stability measures. *IMF Working Paper No.* 09/4.
- Standard, Poor's, 2001. Rating methodology: Evaluating the issuer. <http://www.standardandpoors.com/>.
- Sundaresan, S., Wang, Z., 2009. Y2K and liquidity premium in treasury bond markets. *Rev. Financ. Stud.* 22 (3), 1021–1056.
- Vasicek, O., 1997. *The Loan Loss Distribution*. KMV Corporation.